MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

GATE, NAND, DIGITAL, KEY CODE BUB

INACTIVE FOR NEW DESIGN AFTER DATE OF THIS SPECIFICATION

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic NAND gate module.

1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 1.70 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be BDB.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 75 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/115-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage ($V_{CC}$) of 5.0 ±0.5 Vdc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 Vdc ±1 percent.

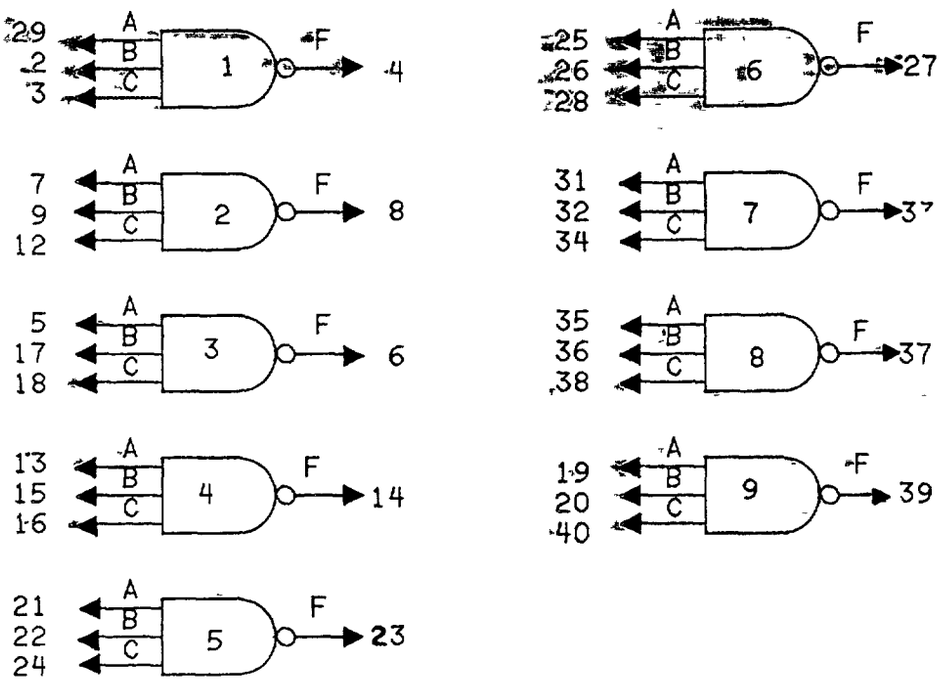
4. QUALITY ASSURANCE PROVISIONS

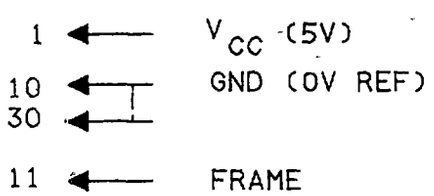
4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

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NOTES:
1. A,B,C = Gate inputs.
2. F = Gate output.
3. Dashed line denotes optional connection internal to module.

FIGURE 1. BDB module function diagram, NAND gate.



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TABLE I. Module electrical requirements.

| Test condition | Parameter | Ref test table | Test type 1/-3/ | Acceptance requirements 25°±5°C ambient temperature | | | | Operating temperature 0° to 60°C | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | Initial | | End-of-life | | Initial | | End-of-life | | |
| | | | | Min | Max | Min | Max | Min | Max | Min | Max | |
| A1-A27 | Zener applied potential, condition A | III | S | 3.8 | 5.5 | 3.5 | 5.5 | 3.8 | 5.5 | 3.5 | 5.5 | V |
| B1-B9 | Zener applied potential, condition B | III | S | 0.00 | 0.40 | 0.00 | 0.44 | 0.00 | 0.42 | 0.00 | 0.46 | V |
| C1-C9 | $V_{OUT}$ (low loaded) | IV | S | 0.0 | 1.1 | 0.0 | 1.1 | 0.0 | 1.1 | 0.0 | 1.1 | V |
| D1 | Isolation | V | S | 10 | ∞ | 1 | ∞ | 10 | ∞ | 1 | ∞ | MΩ |
| D2 | Continuity | V | S | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | Ω |
| E1-E27 | Output voltage, high level | VI | X | 3.8 | 5.5 | 3.5 | 5.5 | 3.8 | 5.5 | 3.5 | 5.5 | V |
| F1-F9 | Output voltage, low level | VI | X | 0.0 | 0.4 | 0.00 | 0.44 | 0.00 | 0.42 | 0.00 | 0.46 | V |
| G1-G9 | Output short circuit current | VII | X | 0.70 | 1.31 | 0.70 | 1.31 | 0.70 | 1.31 | 0.70 | 1.31 | mA |
| H1-H27 | Input current, low level | VIII | X | 0.2 | 1.6 | 0.2 | 1.6 | 0.2 | 1.6 | 0.2 | 1.6 | mA |
| J1-J54 | Input current, high level | VIII | X | 0 | 2 | 0 | 4 | 0 | 3 | 0 | 5 | µA |
| K1-K27 | Input capacitance | IX | Q | 2 | 16 | 2 | 16 | 2 | 16 | 2 | 16 | pF |
| L1 | Power supply current, high level | X | S | 0.0 | 24.8 | 0.0 | 24.8 | 0.0 | 24.8 | 0.0 | 24.8 | mA |
| L2 | Power supply current, low level | X | S | 0.0 | 39.6 | 0.0 | 39.6 | 0.0 | 39.6 | 0.0 | 39.6 | mA |
| M | Power supply transient rise time | XI | S | 36 | 148 | 36 | 148 | 36 | 148 | 36 | 148 | µs |
| N1-N9 | Output pulse amplitude, high level | XII | S | 3.8 | 5.5 | 3.5 | 5.5 | 3.8 | 5.5 | 3.5 | 5.5 | V |
| N10-N18 | Output pulse amplitude, low level | XII | S | 0.0 | 0.4 | 0.00 | 0.44 | 0.00 | 0.42 | 0.00 | 0.46 | V |
| P1-P9 | Transition time ($t_{THL}$) | XIII | S | 6 | 24 | 6 | 24 | 6 | 24 | 6 | 24 | ns |
| P10-P18 | Transition time ($t_{TLH}$) | XIII | S | 60 | 160 | 60 | 160 | 60 | 160 | 60 | 160 | ns |

See footnotes at end of table.



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TABLE I. Module electrical requirements - Continued.

| Test condition | Parameter | Ref test table | Test type 1/-3/ | Acceptance requirements 25°±5°C ambient temperature | | | | Operating temperature 0° to 60°C | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | Initial | | End-of-life | | Initial | | End-of-life | | |
| | | | | Min | Max | Min | Max | Min | Max | Min | Max | |
| Q1-Q9 | Propagation delay time ($t_{PLH}$) | XIV | S | 20 | 75 | 20 | 75 | 20 | 75 | 20 | 75 | ns |
| Q10-Q18 | Propagation delay time ($t_{PHL}$) | XIV | S | 6 | 40 | 6 | 40 | 6 | 40 | 6 | 40 | ns |
| R1-R9 | Output crosstalk, high level | XV | Q | 0.0 | 0.8 | 0.0 | 0.8 | 0.0 | 0.8 | 0.0 | 0.8 | V |
| R10-R18 | Output crosstalk, low level | XV | Q | 0.0 | 0.8 | 0.0 | 0.8 | 0.0 | 0.8 | 0.0 | 0.8 | V |

1/ "X" parameters shall be 100 percent tested during module acceptance tests.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.



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4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through R shall be performed in any order with the following exceptions. If test conditions A, B, C, or D are part of a sequence of tests, the order of testing shall be as follows: A, B, C, D where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through R in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions E through J and L through R shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests E through J and L through R shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests E through R shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Cross reference of classification. Military Specification MIL-M-28787/115 supersedes Department of the Navy, Naval Ordnance Systems Command Drawing No. 10001-2658635D dated 15 March 1971. Part number M28787/115-1 supersedes part numbers 2658635 and M28787/115. These parts are interchangeable.

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6.4 Module description. The following electrical parameters reflect the 0° to 60°C end-of-life limits from table I.

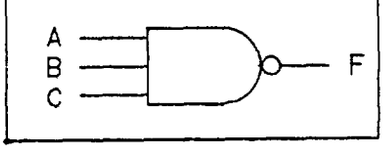
| | |
|---|---|
| Power dissipation | |
| (all outputs at low level) | 218 mW maximum (outputs not loaded) |
| (all outputs at high level) | 136 mW maximum (outputs not loaded) |
| Supply voltage | 5.0 V ±10 percent |
| Input voltage | |
| High level | 1.8 V minimum, $V_{CC}$ maximum |
| Low level | -0.5 V minimum, 1.2 V maximum |
| Input current | |
| High level | 5.0 µA maximum |
| Low level | -1.6 mA maximum |
| Output voltage | |
| High level | 3.5 V minimum, $V_{CC}$ maximum |
| Low level | 0.0 V minimum, 0.46 V maximum |
| Output drive current | |
| High level | -60 µA maximum |
| Low level | 11.2 mA maximum |

6.5 Module function. This module consists of nine 3-input NAND gates with independent inputs and outputs.

6.5.1 Input to output relationship. Each NAND gate has the input to output relationship for positive logic as shown in table II.

TABLE II. Input to output relationship for each NAND gate. 1/ - 5/

| INPUTS | | | OUTPUT |
|---|---|---|---|
| A | B | C | F |
| L | X | X | H |
| X | L | X | H |
| X | X | L | H |
| H | H | H | L |




1/ A, B, C - Gate inputs.
2/ F - Gate output.
3/ H = High level.
4/ L = Low level.
5/ X = Don't care.

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6.5.2 Paralleling of outputs. The paralleling of outputs is permissible with a reduction in low level fan out calculated from the following expression:

Reduced low level fan out = Maximum fan out - (5/6) (n-1)

where maximum fan out = 7 SLL (Standard Logic Loads, low level) and n = the number of outputs connected in parallel.

The increase in high level fan out is calculated from the following expression:

Increased high level fan out = (Maximum fan out) (n)

where maximum fan out = 12 SLL (Standard Logic Loads, high level) and n = the number of outputs connected in parallel.

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

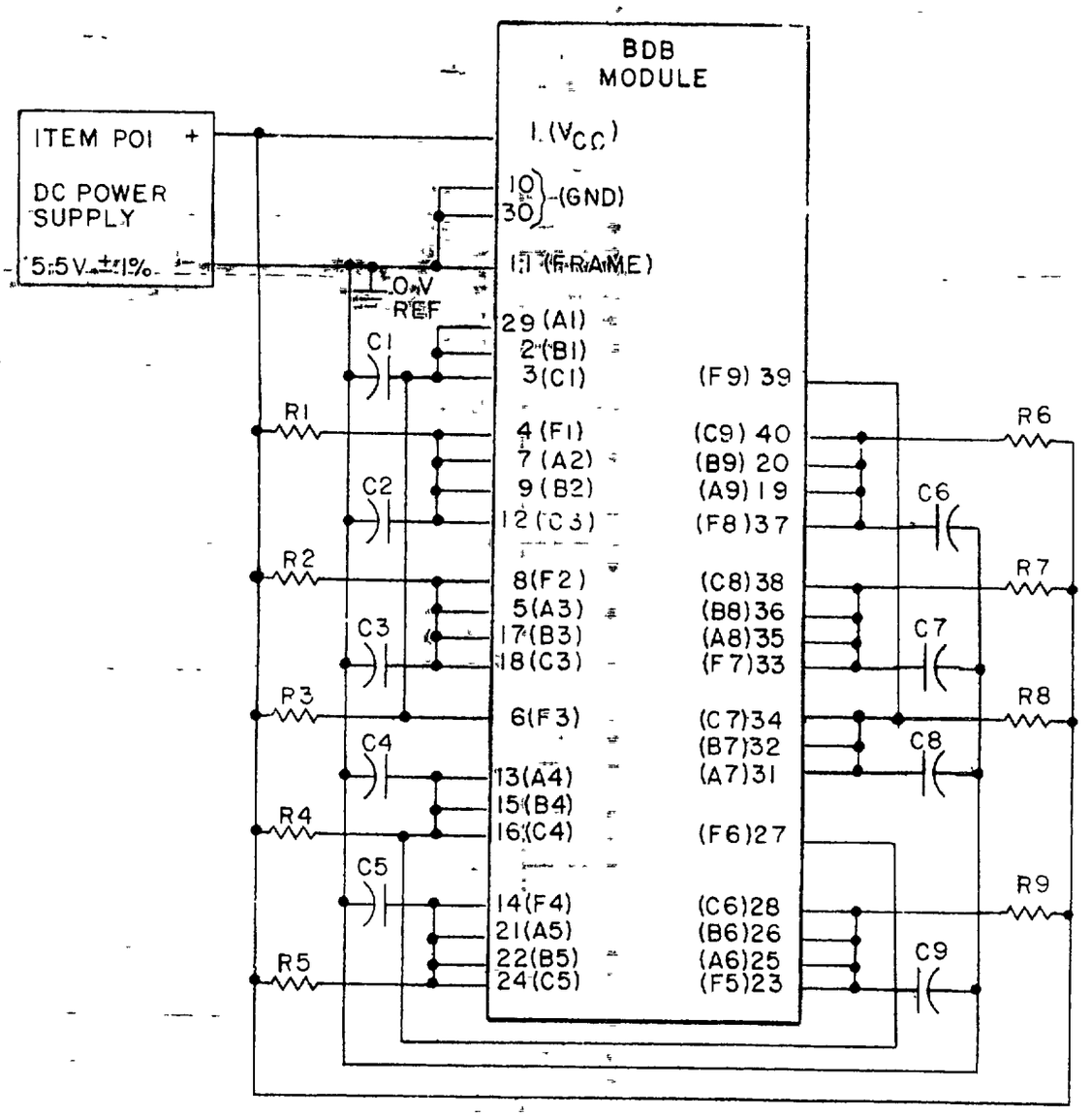
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Preparing activity:
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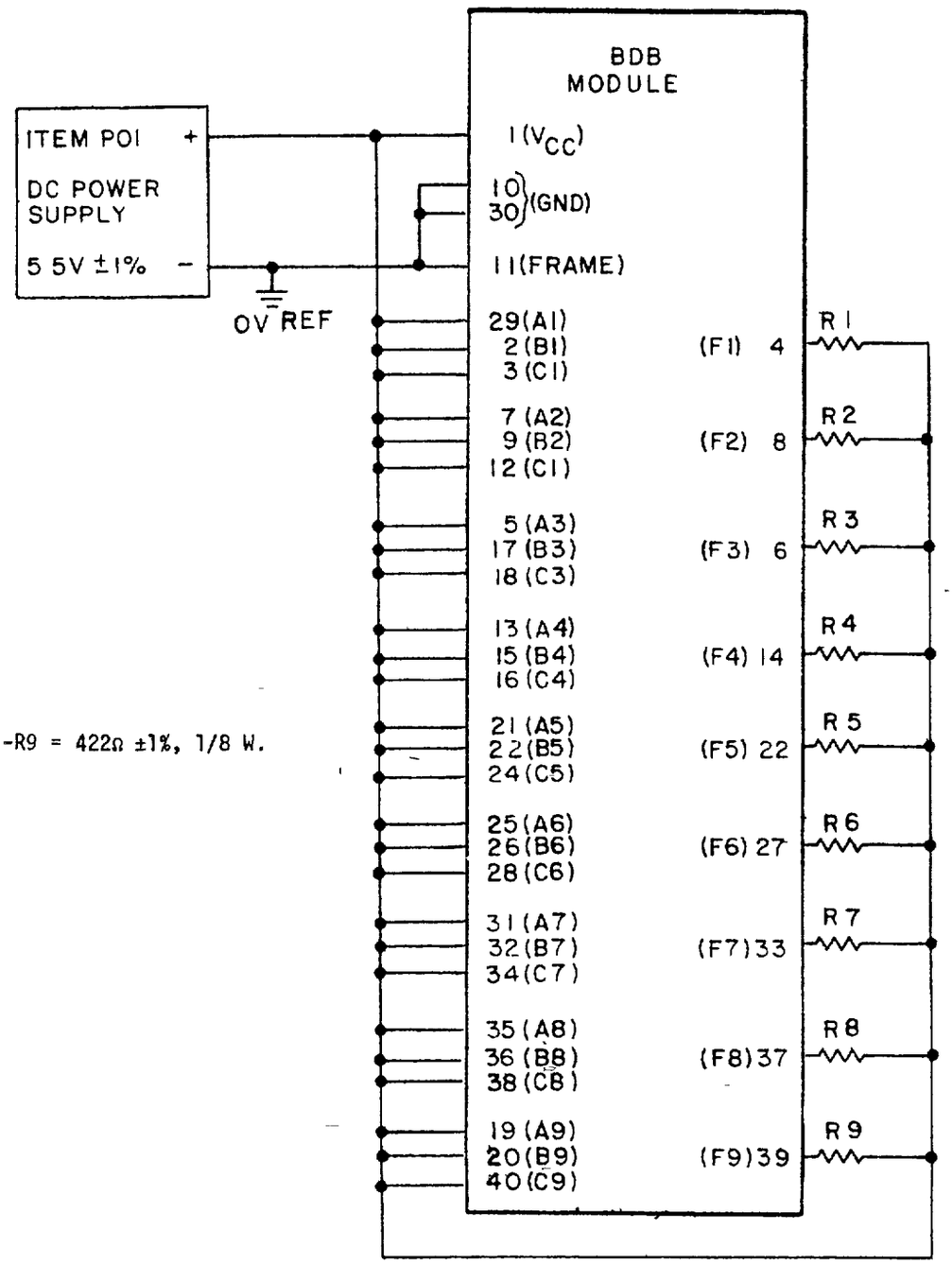
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NOTES: 1. R1-R9 = 422Ω ±1%, 1/8 W.
2. C1-C9 = 220 pF ±10%.

FIGURE 2. Operating temperature and life test circuit.

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NOTE R1-R9 = 422Ω ±1%, 1/8 W.

FIGURE 3. Component temperature test circuit.



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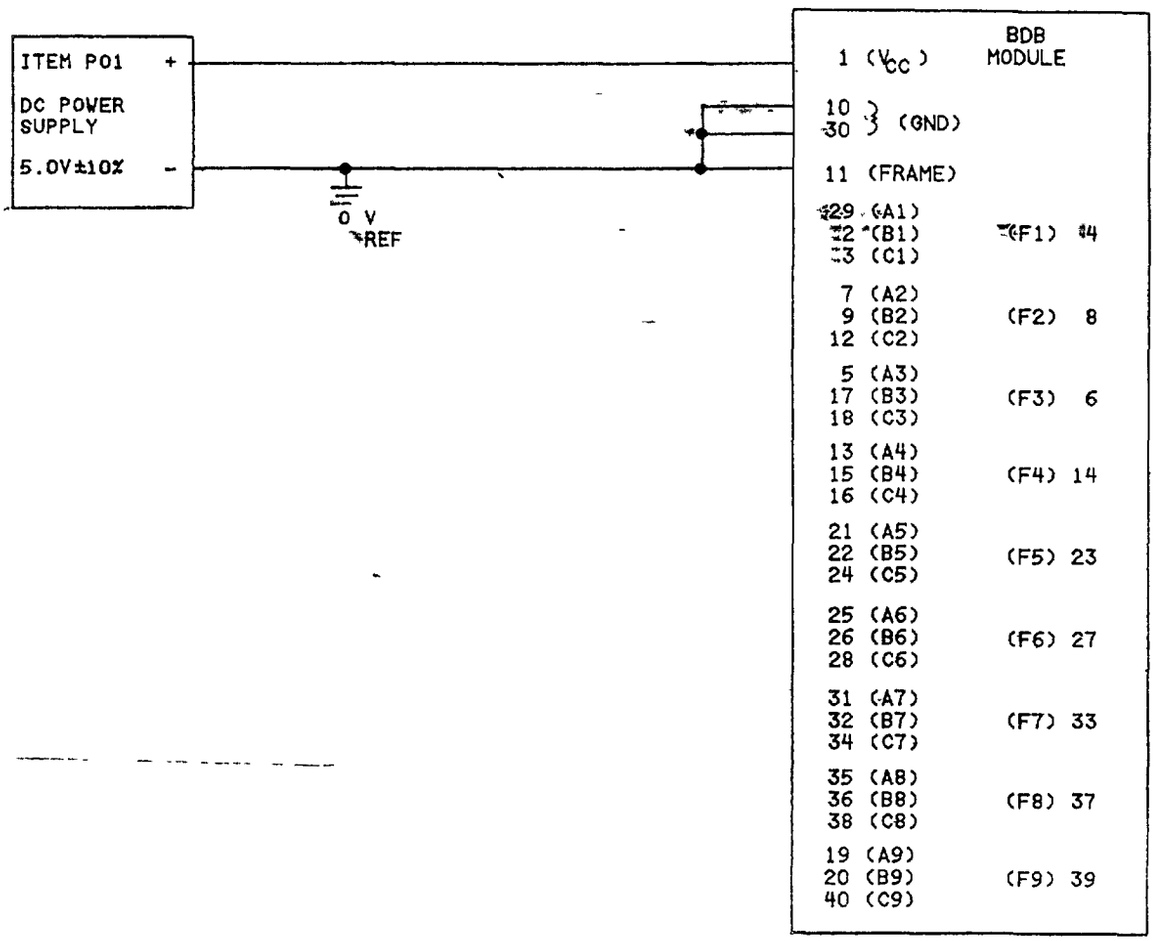



FIGURE 4. Vibration test circuit.



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TABLE III. Zener applied potential test.

| Test condition | Parameter | Connect | | | | | Initial limits 25°±5°C ambient | | Units | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | D to pin | E to pin | Min | Max | | |
| A1 | F1 Zener applied potential, condition A | Open | 4 | 2,3 | 1 | 29 | 3.8 | 5.5 | V | 5,6 |
| A2 | F1 | | 4 | 29,3 | | 2 | | | | |
| A3 | F1 | | 4 | 29,2 | | 3 | | | | |
| A4 | F2 | | 8 | 9,12 | | 7 | | | | |
| A5 | F2 | | 8 | 7,12 | | 9 | | | | |
| A6 | F2 | | 8 | 7,9 | | 12 | | | | |
| A7 | F3 | | 6 | 17,18 | | 5 | | | | |
| A8 | F3 | | 6 | 5,18 | | 17 | | | | |
| A9 | F3 | | 6 | 5,17 | | 18 | | | | |
| A10 | F4 | | 14 | 15,16 | | 13 | | | | |
| A11 | F4 | | 14 | 13,16 | | 15 | | | | |
| A12 | F4 | | 14 | 13,15 | | 16 | | | | |
| A13 | F5 | | 23 | 22,24 | | 21 | | | | |
| A14 | F5 | | 23 | 21,24 | | 22 | | | | |
| A15 | F5 | | 23 | 21,22 | | 24 | | | | |
| A16 | F6 | | 27 | 26,28 | | 25 | | | | |
| A17 | F6 | | 27 | 25,28 | | 26 | | | | |
| A18 | F6 | | 27 | 25,26 | | 28 | | | | |
| A19 | F7 | | 33 | 32,34 | | 31 | | | | |
| A20 | F7 | | 33 | 31,34 | | 32 | | | | |
| A21 | F7 | | 33 | 31,32 | | 34 | | | | |





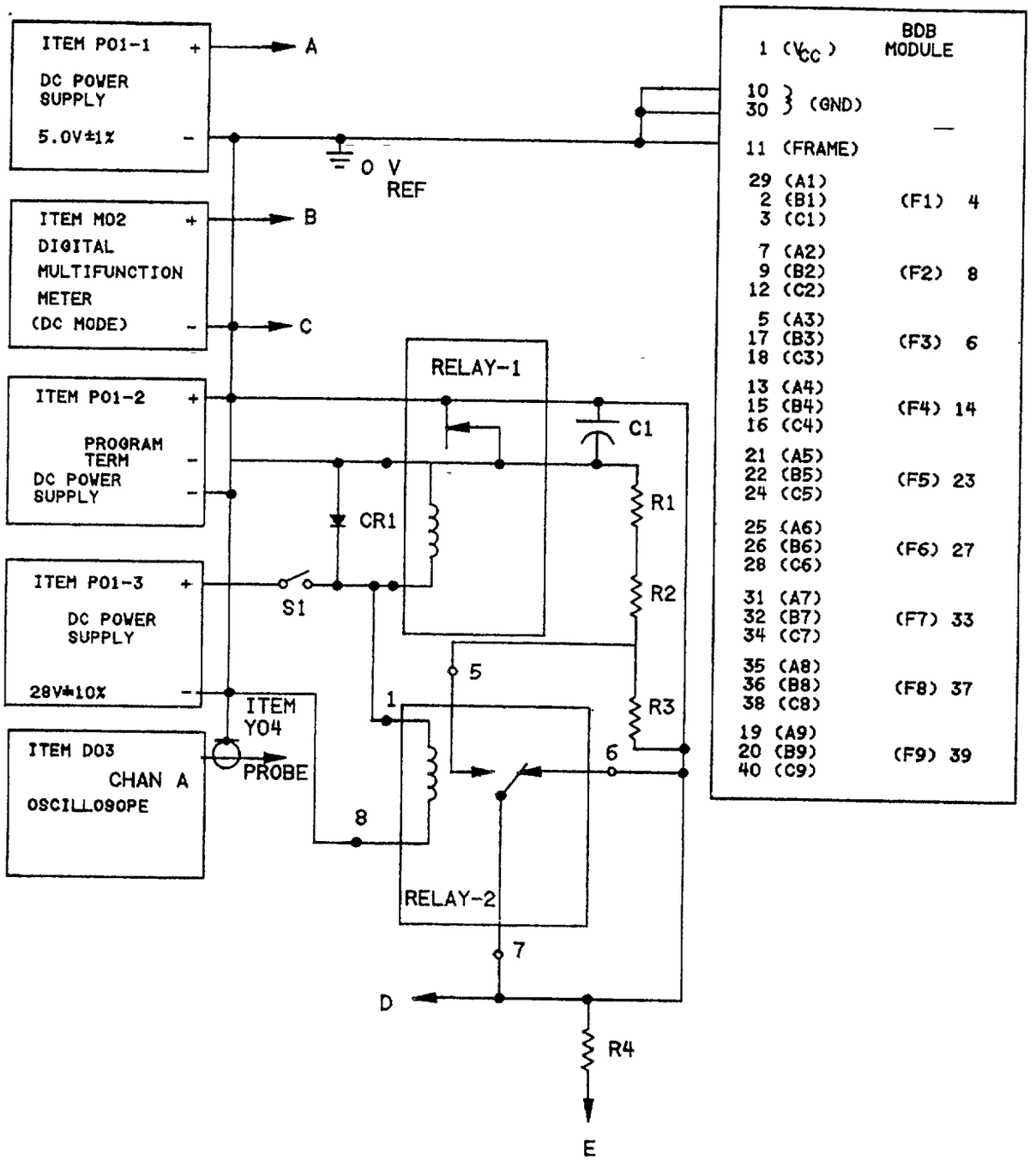
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TABLE III. Zener applied potential test - continued.

| Test condition | Parameter | Connect | | | | | Initial limits 25°±5°C ambient | | Units | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | D to pin | E to pin | Min | Max | | |
| A22 | F8 Zener applied potential, condition A | Open | 37 | 36,38 | 1 | 35 | 3.8 | 5.5 | V | 5,6 |
| A23 | F8 | | 37 | 35,38 | | 36 | | | | |
| A24 | F8 | | 37 | 35,36 | | 38 | | | | |
| A25 | F9 | | 39 | 20,40 | | 19 | | | | |
| A26 | F9 | | 39 | 19,40 | | 20 | | | | |
| A27 | F9 | | 39 | 20,19 | | 40 | | | | |
| B1 | F1 Zener applied potential, condition B | 1 | 4 | Open | Open | 29 | 0.00 | 0.40 | | |
| B2 | F2 | | 8 | | | 7 | | | | |
| B3 | F3 | | 6 | | | 5 | | | | |
| B4 | F4 | | 14 | | | 13 | | | | |
| B5 | F5 | | 23 | | | 21 | | | | |
| B6 | F6 | | 27 | | | 25 | | | | |
| B7 | F7 | | 33 | | | 31 | | | | |
| B8 | F8 | | 37 | | | 35 | | | | |
| B9 | F9 | | 39 | | | 19 | | | | |



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CAUTION:

If these tests are successively repeated, a time interval of 10 seconds minimum between tests shall be allowed to preclude detrimental effects due to module heating.

FIGURE 5. Zener applied potential test circuit for table III.

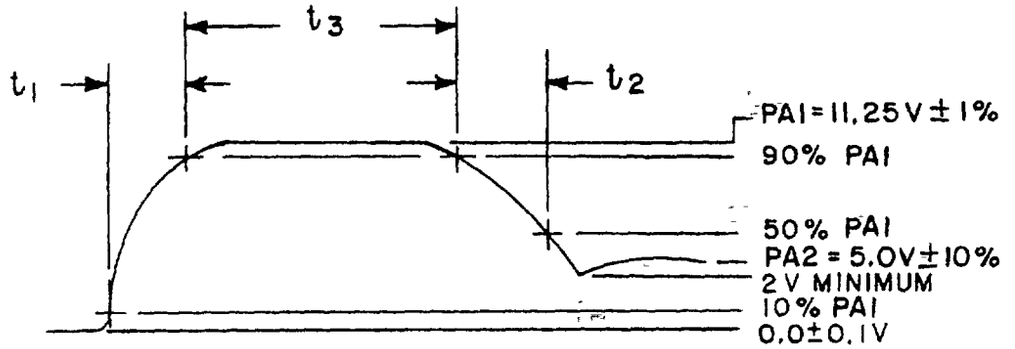


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NOTES.
1. R1 = 178Ω ±1%, 1/8 W.
2. R2 = 301Ω ±1%, 1/8 W.
3. R3 = 649Ω ±1%, 1/8 W.
4. R4 = 1.54Ω ±1%, 1/2 W.
5. C1 = 22 µF ±10%.
6. CR1 = 1N647, or equivalent.
7. Relay-1 = GE 3SAV1256A2, or equivalent.
8. Relay-2 = Agastat 2112, or equivalent.
9. S1 = SPST switch.
10. Resistors R1, R2, R3, and capacitor C1 are programming components for item PO1-2 and are based on a programming constant of 100 ohms per volt.
11. Using item DO3 with item YO4 probe, adjust the time delay relay to obtain the output waveform as specified on figure 6 at pin 7 of the time delay relay after switch S1 is closed.
12. For each test condition, switch S1 is closed and the voltage reading is made a minimum of 2 seconds after the closure of switch S1. At the conclusion of each test condition measurement, switch S1 is returned to the open position.
13. This circuit is drawn showing item PO1-2 with a negative reference terminal (non-commoned programming terminal labeled (-)). This circuit, as shown, will not operate properly if item PO1-2 has a positive non-commoned programming terminal. NOTE: Load current must not pass through the sensing network.

FIGURE 5. Zener applied potential test circuit for table III - Continued.

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NOTES:
1. t1 = 100 +100, -50 ms.
2. t2 = 100 +400, -50 ms.
3. t3 = 0.5s ±10%.

FIGURE 6. Zener applied potential test circuit waveform for table III.



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TABLE IV. $V_{OUT}$ (low-loaded) test.

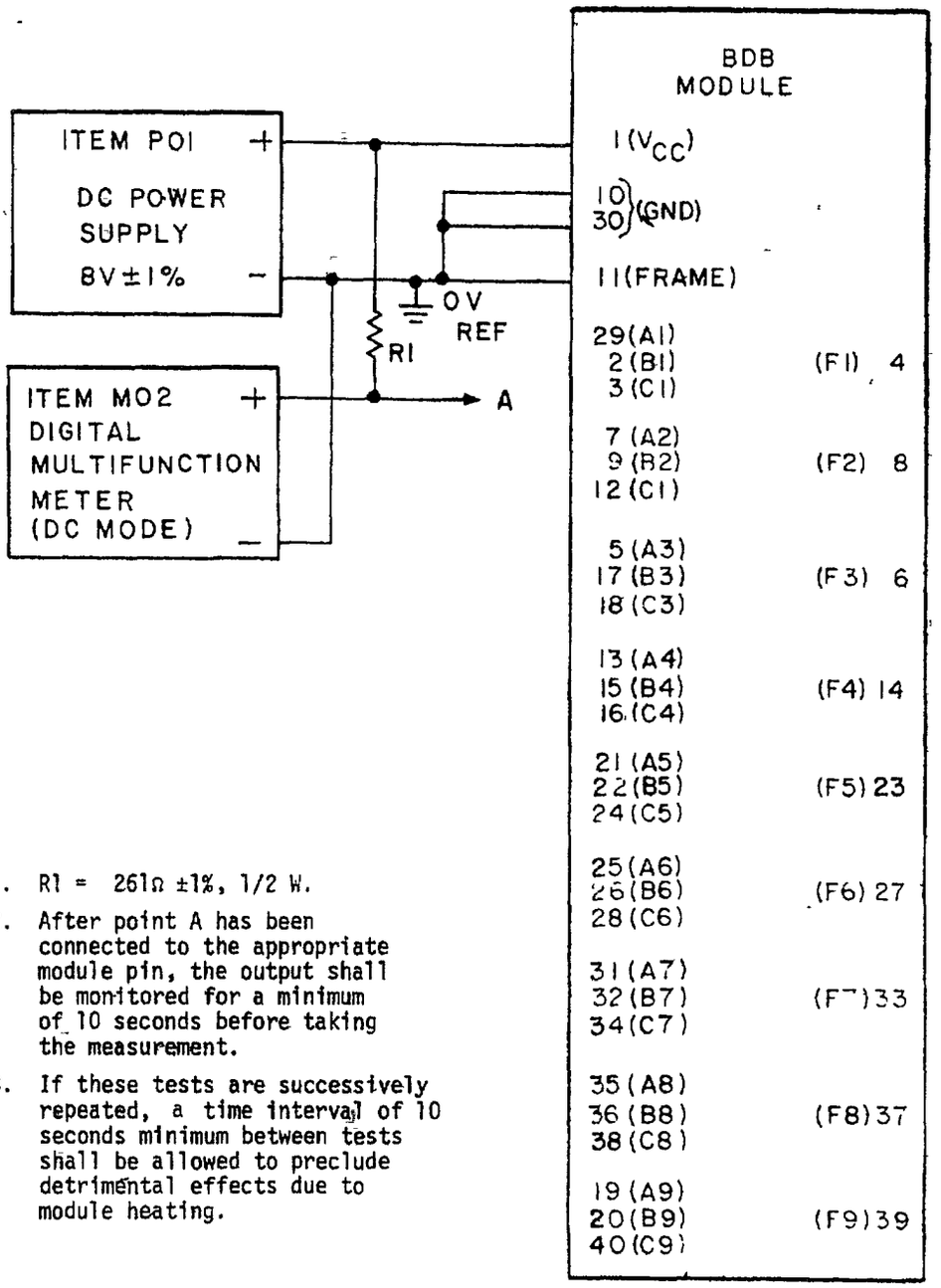
| Test condition | Parameter | Connect A to pin | Initial limits 25° ±5°C ambient Min | Initial limits 25° ±5°C ambient Max | Units | Test figure |
|---|---|---|---|---|---|---|
| C1 | F1 $V_{OUT}$ (low loaded) | 4 | 0.0 | 1.1 | V | 7 |
| C2 | F2 | 8 | | | | |
| C3 | F3 | 6 | | | | |
| C4 | F4 | 14 | | | | |
| C5 | F5 | 23 | | | | |
| C6 | F6 | 27 | | | | |
| C7 | F7 | 33 | | | | |
| C8 | F8 | 37 | | | | |
| C9 | F9 | 39 | | | | |

TABLE V. Isolation and continuity test.

| Test condition | Parameter | Connect A to pin | Connect B to pin | Connect C to pin | Connect D to pin | Initial limits 25° ±5°C ambient Min | Initial limits 25° ±5°C ambient Max | Units | Test figure |
|---|---|---|---|---|---|---|---|---|---|
| D1 | Isolation | 11 | all other pins | Open | Open | 10 | ∞ | MΩ | 8 |
| D2 | Continuity | Open | Open | 11 | Frame | 0 | 1 | Ω | 8 |



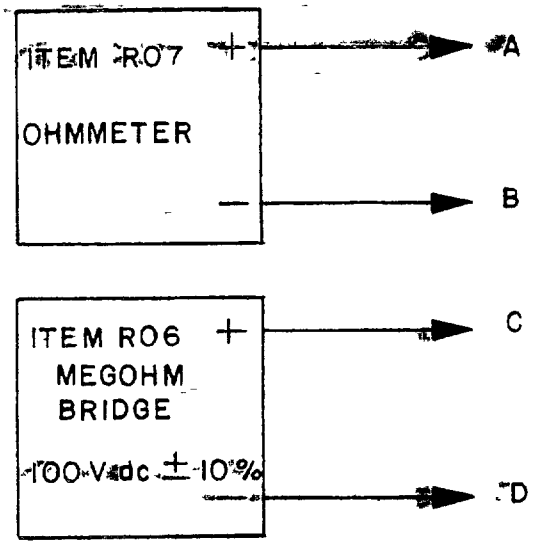
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NOTES:
1. R1 = 261Ω ±1%, 1/2 W.
2. After point A has been connected to the appropriate module pin, the output shall be monitored for a minimum of 10 seconds before taking the measurement.
3. If these tests are successively repeated, a time interval of 10 seconds minimum between tests shall be allowed to preclude detrimental effects due to module heating.

FIGURE 7. $V_{OUT}$ (low-loaded) test circuit for table IV.

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WARNING:

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE: For test condition D2 penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

*B,D,B MODULE

| | |
|---|---|
| 1 ($V_{CC}$) | |
| 10, 30 (GND) | |
| 11 (FRAME) | |
| 29 (A1), 2 (B1), 3 (C1) | (F1) 4 |
| 7 (A2), 9 (B2), 12 (C1) | (F2) #8 |
| 5 (A3), 17 (B3), 18 (C3) | (F3) 6 |
| 13 (A4), 15 (B4), 16 (C4) | (F4) 14 |
| 21 (A5), 22 (B5), 24 (C5) | (F5) 23 |
| 25 (A6), 26 (B6), 28 (C6) | (F6) 27 |
| 31 (A7), 32 (B7), 34 (C7) | (F7) 33 |
| 35 (A8), 36 (B8), 38 (C8) | (F8) 37 |
| 19 (A9), 20 (B9), 40 (C9) | (F9) 39 |

FIGURE 8. Isolation and continuity test circuit for table V.



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TABLE VI Output voltage (high-level and low-level) test.

| Test condition | Parameter | | Connect | | | | Initial limits 25°±5°C ambient | | Units | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | A to pin | B to pin | C to pin | D to pin | Min | Max | | |
| E1 | F1 | Output voltage, high level | 29 | Open | 4 | 4 | 3.8 | 5.5 | V | 9 |
| E2 | F1 | | 2 | | 4 | 4 | | | | |
| E3 | F1 | | 3 | | 4 | 4 | | | | |
| E4 | F2 | | 7 | | 8 | 8 | | | | |
| E5 | F2 | | 9 | | 8 | 8 | | | | |
| E6 | F2 | | 12 | | 8 | 8 | | | | |
| E7 | F3 | | 5 | | 6 | 6 | | | | |
| E8 | F3 | | 17 | | 6 | 6 | | | | |
| E9 | F3 | | 18 | | 6 | 6 | | | | |
| E10 | F4 | | 13 | | 14 | 14 | | | | |
| E11 | F4 | | 15 | | 14 | 14 | | | | |
| E12 | F4 | | 16 | | 14 | 14 | | | | |
| E13 | F5 | | 21 | | 23 | 23 | | | | |
| E14 | F5 | | 22 | | 23 | 23 | | | | |
| E15 | F5 | | 24 | | 23 | 23 | | | | |
| E16 | F6 | | 25 | | 27 | 27 | | | | |
| E17 | F6 | | 26 | | 27 | 27 | | | | |
| E18 | F6 | | 28 | | 27 | 27 | | | | |
| E19 | F7 | | 31 | | 33 | 33 | | | | |



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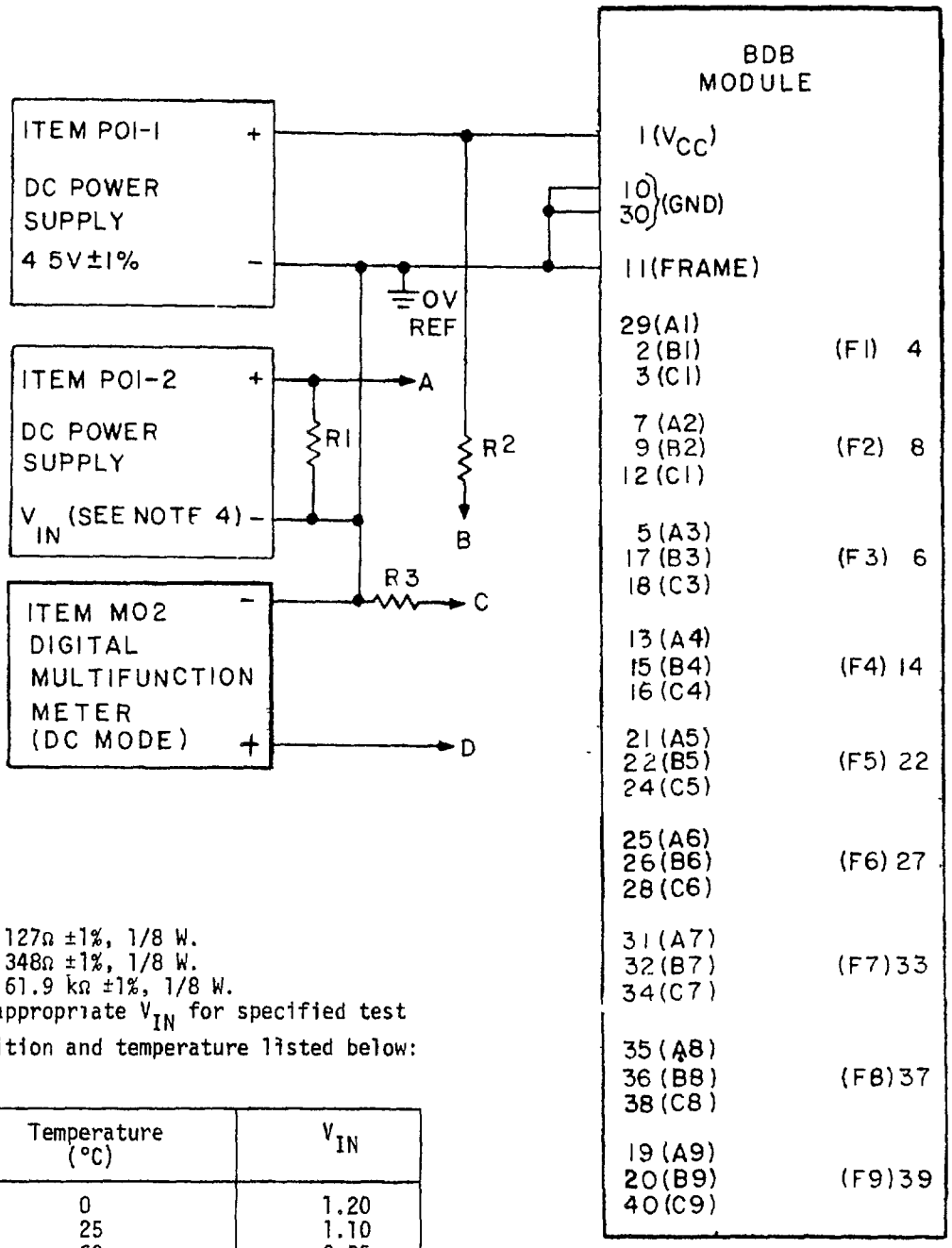
TABLE VI. Output voltage (high-level and low-level) test - Continued.

| Test condition | Parameter | Connect | | | | Initial limits 25°±5°C ambient | | Units | Test figure |
|---|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | D to pin | Min | Max | | |
| E20 | F7 Output voltage, high level | 32 | Open | 33 | 33 | 3.8 | 5.5 | V | 9 |
| E21 | F7 | 34 | | 33 | 33 | | | | |
| E22 | F8 | 35 | | 37 | 37 | | | | |
| E23 | F8 | 36 | | 37 | 37 | | | | |
| E24 | F8 | 38 | | 37 | 37 | | | | |
| E25 | F9 | 19 | | 39 | 39 | | | | |
| E26 | F9 | 20 | | 39 | 39 | | | | |
| E27 | F9 | 40 | | 39 | 39 | | | | |
| F1 | F1 Output voltage, low level | 29,2,3 | 4 | Open | 4 | 0.0 | 0.4 | | |
| F2 | F2 | 7,9,12 | 8 | | 8 | | | | |
| F3 | F3 | 5,17,18 | 6 | | 6 | | | | |
| F4 | F4 | 13,15,16 | 14 | | 14 | | | | |
| F5 | F5 | 21,22,24 | 23 | | 23 | | | | |
| F6 | F6 | 25,26,28 | 27 | | 27 | | | | |
| F7 | F7 | 31,32,34 | 33 | | 33 | | | | |
| F8 | F8 | 35,36,38 | 37 | | 37 | | | | |
| F9 | F9 | 19,20,40 | 39 | | 39 | | | | |





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NOTES:
1. R1 = 127Ω ±1%, 1/8 W.
2. R2 = 348Ω ±1%, 1/8 W.
3. R3 = 61.9 kΩ ±1%, 1/8 W.
4. Use appropriate $V_{IN}$ for specified test condition and temperature listed below:

| Test condition | Temperature (°C) | $V_{IN}$ |
|---|---|---|
| E1-E27 | 0 | 1.20 |
| | 25 | 1.10 |
| | 60 | 0.95 |
| | 80 | 0.80 |
| F1-F9 | 0 | 2.0 |
| | 25 | 1.9 |
| | 60 | 1.8 |
| | 80 | 1.8 |

0°, 60°, and 80°C refer to operating temperatures (module frame); 25°C refers to ambient temperature. For 80°C use the 0° to 60° end-of-life limits.

FIGURE 9. Output voltage (high-level and low-level) test.

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TABLE VII. Output short circuit current test.

| Test condition | Parameter | Connect | | | Initial limits 25°±5°C ambient | | Units | Test figure |
|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | Min | Max | | |
| G1 | F1 Output short circuit current | Open | 4 | 29,2,3 | 0.70 | 1.31 | mA | 10 |
| G2 | F2 | | 8 | 7,9,12 | | | | |
| G3 | F3 | | 6 | 5,17,18 | | | | |
| G4 | F4 | | 14 | 13,15,16 | | | | |
| G5 | F5 | | 23 | 21,22,24 | | | | |
| G6 | F6 | | 27 | 25,26,28 | | | | |
| G7 | F7 | | 33 | 31,32,34 | | | | |
| G8 | F8 | | 37 | 35,36,38 | | | | |
| G9 | F9 | | 39 | 19,20,40 | | | | |





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TABLE VIII. Input current (low-level and high-level) test.

| Test condition | | Parameter | Connect A to pin | Connect B to pin | Connect C to pin | Initial limits 25°±5°C ambient Min | Initial limits 25°±5°C ambient Max | Units | Test figure |
|---|---|---|---|---|---|---|---|---|---|
| H1 | A1 | Input current, low level | Open | 29 | Open | 0.2 | 1.6 | mA | 10 |
| H2 | B1 | | | 2 | | | | | |
| H3 | C1 | | | 3 | | | | | |
| H4 | A2 | | | 7 | | | | | |
| H5 | B2 | | | 9 | | | | | |
| H6 | C2 | | | 12 | | | | | |
| H7 | A3 | | | 5 | | | | | |
| H8 | B3 | | | 17 | | | | | |
| H9 | C3 | | | 18 | | | | | |
| H10 | A4 | | | 13 | | | | | |
| H11 | B4 | | | 15 | | | | | |
| H12 | C4 | | | 16 | | | | | |
| H13 | A5 | | | 21 | | | | | |
| H14 | B5 | | | 22 | | | | | |
| H15 | C5 | | | 24 | | | | | |
| H16 | A6 | | | 25 | | | | | |
| H17 | B6 | | | 26 | | | | | |
| H18 | C6 | | | 28 | | | | | |
| H19 | A7 | | | 31 | | | | | |
| H20 | B7 | | | 32 | | | | | |
| H21 | C7 | | | 34 | | | | | |

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TABLE VIII. Input current (low-level and high-level) test - Continued.

| Test condition | Parameter | | Connect | | | Initial limits 25°±5°C ambient | | Units | Test figure |
|---|---|---|---|---|---|---|---|---|---|
| | | | A to pin | B to pin | C to pin | Min | Max | | |
| H22 | A8 | Input current, low level | Open | 35 | Open | 0.2 | 1.6 | mA | 10 |
| H23 | B8 | | | 36 | | | | | |
| H24 | C8 | | | 38 | | | | | |
| H25 | A9 | | | 19 | | | | | |
| H26 | B9 | | | 20 | | | | | |
| H27 | C9 | | | 40 | | | | | |
| J1 | A1 | Input current, high level | 29 | Open | 2 | 0 | 2 | µA | |
| J2 | A1 | | 29 | | 3 | | | | |
| J3 | B1 | | 2 | | 29 | | | | |
| J4 | B1 | | 2 | | 3 | | | | |
| J5 | C1 | | 3 | | 29 | | | | |
| J6 | C1 | | 3 | | 2 | | | | |
| J7 | A2 | | 7 | | 9 | | | | |
| J8 | A2 | | 7 | | 12 | | | | |
| J9 | B2 | | 9 | | 7 | | | | |
| J10 | B2 | | 9 | | 12 | | | | |
| J11 | C2 | | 12 | | 7 | | | | |
| J12 | C2 | | 12 | | 9 | | | | |





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TABLE VIII. Input current (low-level and high-level) test - Continued.

| Test condition | Parameter | Connect | | | Initial limits 25°±5°C ambient | | Units | Test figure |
|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | Min | Max | | |
| J13 | A3 Input current, high level | 5 | Open | 17 | 0 | 2 | µA | 10 |
| J14 | A3 | 5 | | 18 | | | | |
| J15 | B3 | 17 | | 5 | | | | |
| J16 | B3 | 17 | | 18 | | | | |
| J17 | C3 | 18 | | 5 | | | | |
| J18 | C3 | 18 | | 17 | | | | |
| J19 | A4 | 13 | | 15 | | | | |
| J20 | A4 | 13 | | 16 | | | | |
| J21 | B4 | 15 | | 13 | | | | |
| J22 | B4 | 15 | | 16 | | | | |
| J23 | C4 | 16 | | 13 | | | | |
| J24 | C4 | 16 | | 15 | | | | |
| J25 | A5 | 21 | | 22 | | | | |
| J26 | A5 | 21 | | 24 | | | | |
| J27 | B5 | 22 | | 21 | | | | |
| J28 | B5 | 22 | | 24 | | | | |
| J29 | C5 | 24 | | 21 | | | | |
| J30 | C5 | 24 | | 22 | | | | |
| J31 | A6 | 25 | | 26 | | | | |
| J32 | A6 | 25 | | 28 | | | | |



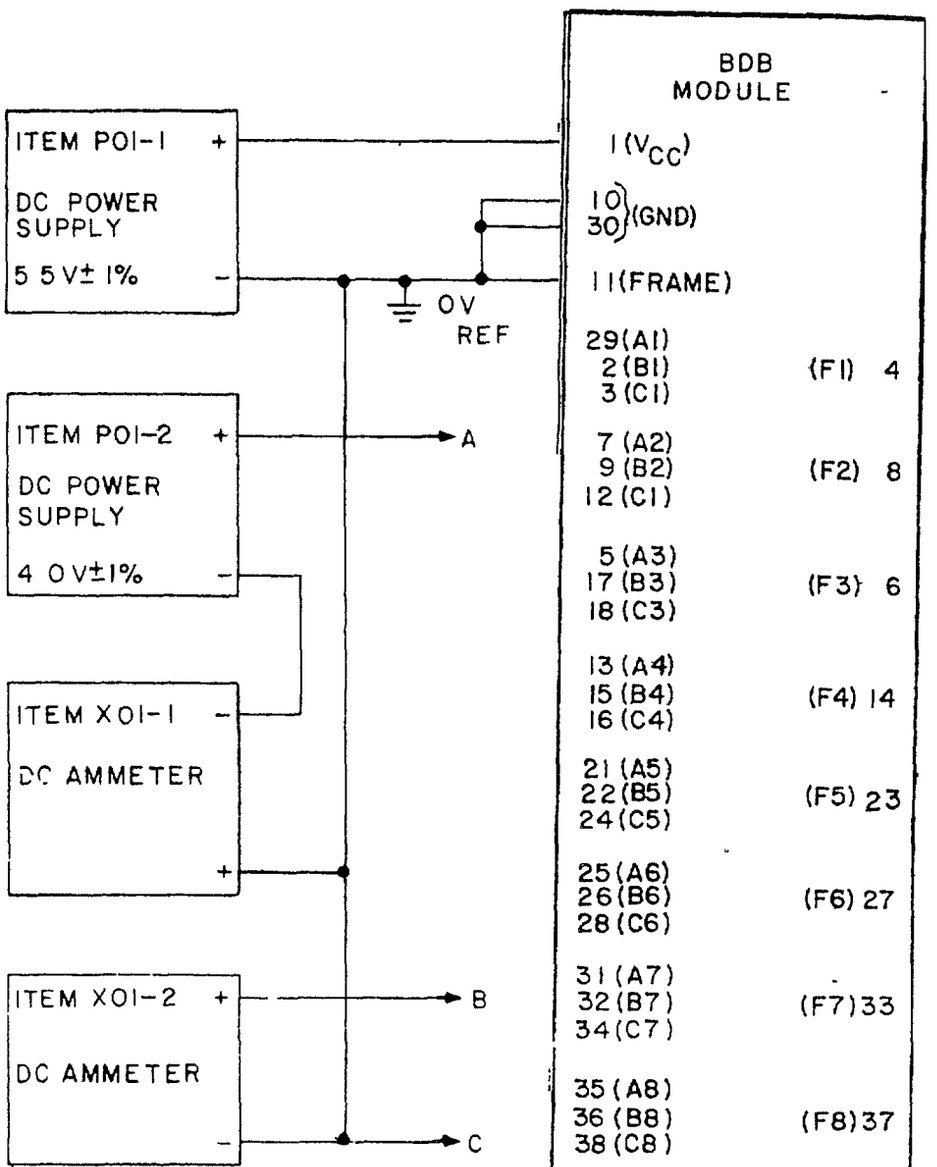
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TABLE VIII. Input current (low-level and high-level) test - Continued.

| Test condition | Parameter | Connect | | | Initial limits 25°±5°C ambient | | Units | Test figure |
|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | Min | Max | | |
| J33 | B6 Input current, high level | 26 | Open | 25 | 0 | 2 | µA | 10 |
| J34 | B6 | 26 | | 28 | | | | |
| J35 | C6 | 28 | | 25 | | | | |
| J36 | C6 | 28 | | 26 | | | | |
| J37 | A7 | 31 | | 32 | | | | |
| J38 | A7 | 31 | | 34 | | | | |
| J39 | B7 | 32 | | 31 | | | | |
| J40 | B7 | 32 | | 34 | | | | |
| J41 | C7 | 34 | | 31 | | | | |
| J42 | C7 | 34 | | 32 | | | | |
| J43 | A8 | 35 | | 36 | | | | |
| J44 | A8 | 35 | | 38 | | | | |
| J45 | B8 | 36 | | 35 | | | | |
| J46 | B8 | 36 | | 38 | | | | |
| J47 | C8 | 38 | | 35 | | | | |
| J48 | C8 | 38 | | 36 | | | | |
| J49 | A9 | 19 | | 20 | | | | |
| J50 | A9 | 19 | | 40 | | | | |
| J51 | B9 | 20 | | 19 | | | | |
| J52 | B9 | 20 | | 40 | | | | |
| J53 | C9 | 40 | | 19 | | | | |
| J54 | C9 | 40 | | 20 | | | | |



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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

CAUTION

The chassis of item P01-2 shall be isolated from earth ground.

NOTE: Since items X01-1 and X01-2 are not used simultaneously, only one item X01 is required. Use item X01-2 for test conditions G1-G9 and H1-H27 and item X01-1 for test conditions J1-J54.

FIGURE 10. Output short circuit current and input current (low-level and high-level) test circuit for tables VII and VIII.



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TABLE IX. Input capacitance test.

| Test condition | Parameter | Connect A to pin | Initial limits 25° ±5°C ambient, Min | Initial limits 25° ±5°C ambient, Max | Units | Test figure |
|---|---|---|---|---|---|---|
| K1 | A1 Input capacitance | 29 | 2 | 16 | pF | 11 |
| K2 | B1 | 2 | | | | |
| K3 | C1 | 3 | | | | |
| K4 | A2 | 7 | | | | |
| K5 | B2 | 9 | | | | |
| K6 | C2 | 12 | | | | |
| K7 | A3 | 5 | | | | |
| K8 | B3 | 17 | | | | |
| K9 | C3 | 18 | | | | |
| K10 | A4 | 13 | | | | |
| K11 | B4 | 15 | | | | |
| K12 | C4 | 16 | | | | |
| K13 | A5 | 21 | | | | |
| K14 | B5 | 22 | | | | |
| K15 | C5 | 24 | | | | |
| K16 | A6 | 25 | | | | |
| K17 | B6 | 26 | | | | |
| K18 | C6 | 28 | | | | |
| K19 | A7 | 31 | | | | |
| K20 | B7 | 32 | | | | |
| K21 | C7 | 34 | | | | |





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TABLE IX. Input capacitance test - Continued.

| Test condition | Parameter | Connect A to pin | Initial limits 25°±5°C ambient Min | Initial limits 25°±5°C ambient Max | Units | Test figure |
|---|---|---|---|---|---|---|
| K22 | A8 Input capacitance | 35 | 2 | 16 | pF | 11 |
| K23 | B8 | 36 | | | | |
| K24 | C8 | 38 | | | | |
| K25 | A9 | 19 | | | | |
| K26 | B9 | 20 | | | | |
| K27 | C9 | 40 | | | | |

TABLE X. Power supply current (high-level and low-level) test.

| Test condition | Parameter | Connect A to pin | Initial limits 25°±5°C ambient Min | Initial limits 25°±5°C ambient Max | Units | Test figure |
|---|---|---|---|---|---|---|
| L1 | Power supply current, high level | 10 | 0.0 | 24.8 | mA | 12 |
| L2 | Power supply current, low level | 1 | 0.0 | 39.6 | mA | 12 |

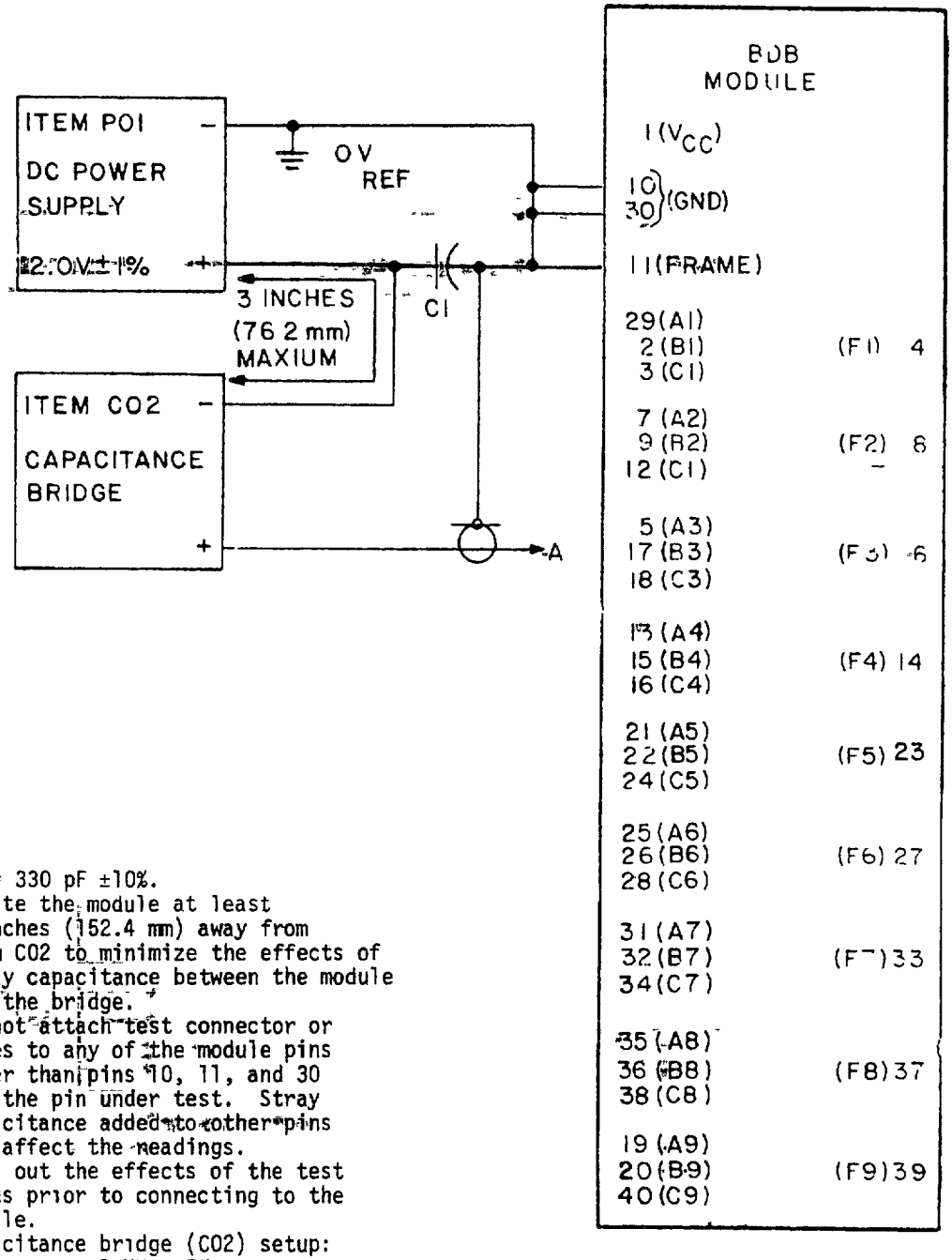
TABLE XI. Power supply transient rise time test.

| Test condition | Parameter | Initial limits 25°±5°C ambient Min | Initial limits 25°±5°C ambient Max | Units | Test figure |
|---|---|---|---|---|---|
| M | Power supply transient rise time | 36 | 148 | µs | 13,14 |





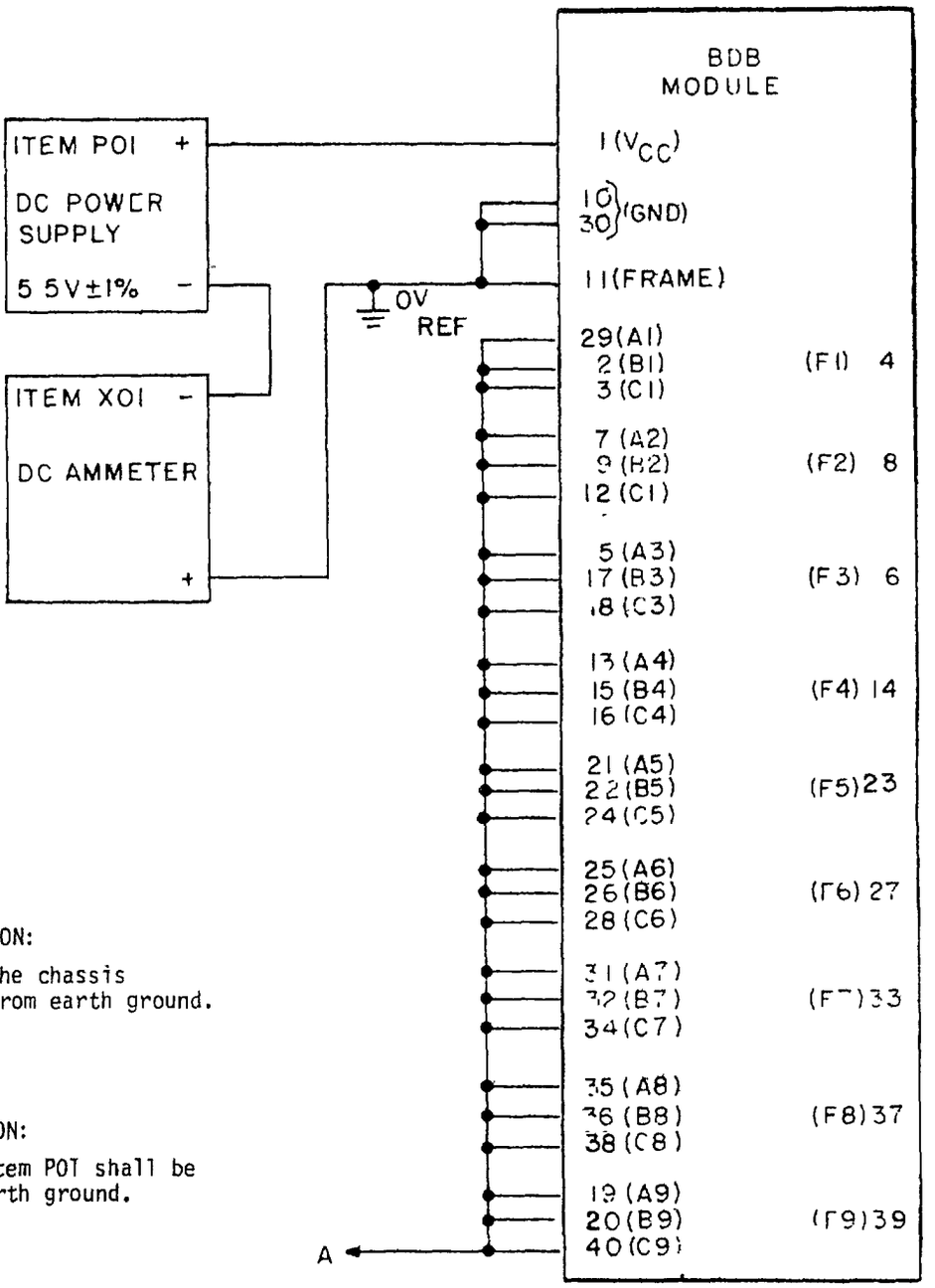
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NOTES:
1. C1 = 330 pF ±10%.
2. Locate the module at least 6 inches (152.4 mm) away from item CO2 to minimize the effects of stray capacitance between the module and the bridge.
3. Do not attach test connector or wires to any of the module pins other than pins 10, 11, and 30 and the pin under test. Stray capacitance added to other pins may affect the readings.
4. Null out the effects of the test leads prior to connecting to the module.
5. Capacitance bridge (CO2) setup: Frequency = 1 MHz ±5%. $V_{pk}$ = 0.5 V maximum.

FIGURE 11. Input capacitance test circuit for table IX.

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

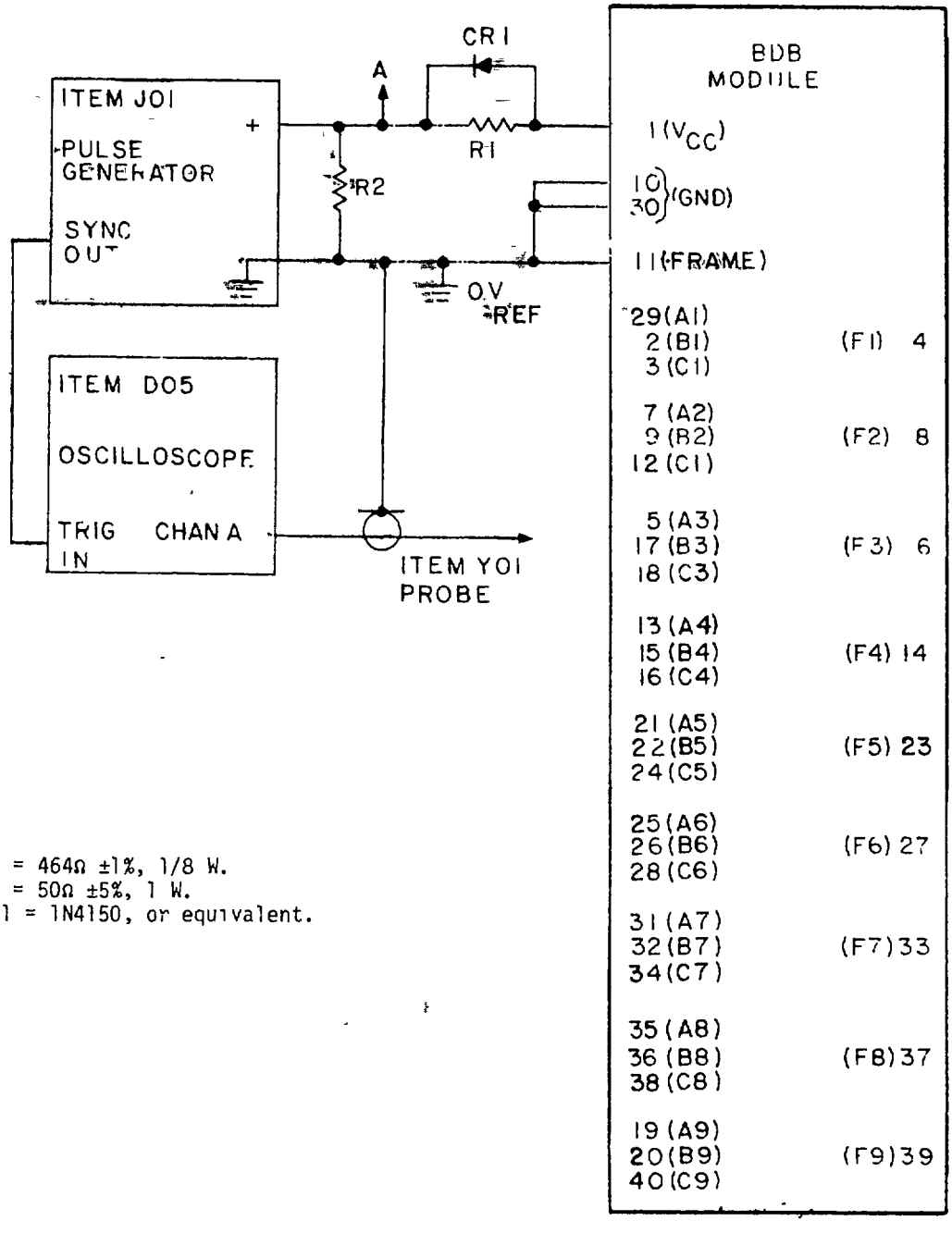
CAUTION:

The chassis of item P01 shall be isolated from earth ground.

FIGURE 12. Power supply current (high-level and low-level) test circuit for table X.



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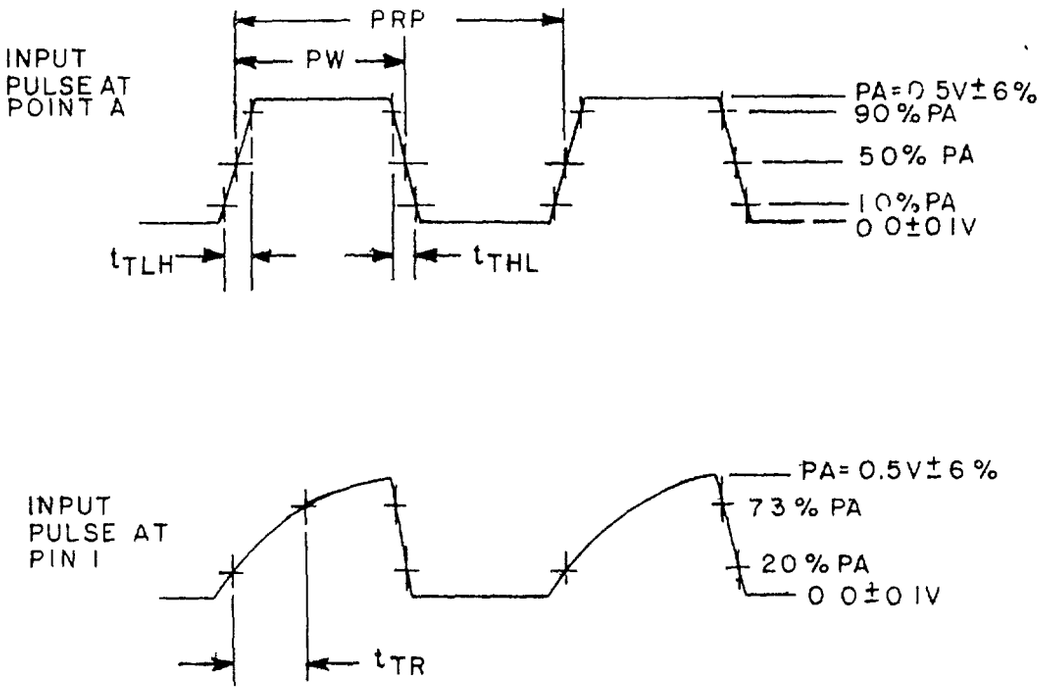



NOTES.
1. R1 = 464Ω ±1%, 1/8 W.
2. R2 = 50Ω ±5%, 1 W.
3. CR1 = 1N4150, or equivalent.

FIGURE 13. Power supply transient rise time test circuit for table XI.



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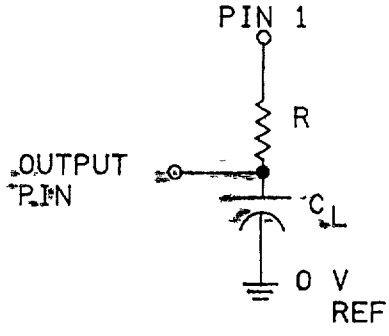

NOTES: 1. Input pulse parameters:

PW = 750 µs ±5%.
PRP = 1.5 ms ±5%.
$t_{THL} = t_{TLH}$ = 50 ns ±10%.

2. Measure transient rise time ($t_{TR}$) at pin 1.

FIGURE 14. Power supply transient rise time test circuit waveforms for table XI.

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LOAD CIRCUIT NUMBER 1.

For test conditions N1-N18, P1-P18 and Q1-Q18:

NOTES: 1. R = 3.83kΩ ±1%, 1/8 W.
2. $C_L$ = 30 ±5 pF (includes stray, probe, and compensating capacitor, as required).
3. With the module unplugged, the oscilloscope probe disconnected, and the 3.83kΩ load resistors disconnected from their respective output pins, use item C02 to measure the capacitance between each test connector output pin and pins 10, 11, and 30 combined. The resistors shall be disconnected from their respective output pins as close to their bodies as practical. The measurements shall be performed at the point of mechanical contact between the module blade pin and the test connector pin. The capacitance shall be measured at 1 MHz ±10%. Reconnect the 3.83kΩ resistors prior to using the test connector for module testing.
4. The loop length from pin 1 to each load resistor and output, including the length of the resistor body, shall not exceed 4.5 inches (114.3 mm).

LOAD CIRCUIT NUMBER 2.

For test conditions R1-R18.

NOTES: 1. R = 422Ω ±1%, 1/8 W.
2. Same as load circuit number 1, note 2.
3. Same ad load circuit number 1, note 3, except use a resistance value of 422Ω in lieu of 3.83kΩ.
4. Same as load circuit number 1, note 4.

FIGURE 15. BDB logic load circuit.

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TABLE XII. Output pulse amplitude (high-level and low-level) test.

| Test condition | Parameter | Connect | | | Initial limits 25°±5°C ambient | | Units | Test figure |
|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | Min | Max | | |
| N1 | F1 Output pulse amplitude, high level | 2 | 29,3 | 4 | 3.8 | 5.5 | V | 16,17 |
| N2 | F2 | 12 | 9,7 | 8 | | | | |
| N3 | F3 | 18 | 17,5 | 6 | | | | |
| N4 | F4 | 16 | 15,13 | 14 | | | | |
| N5 | F5 | 21 | 22,24 | 23 | | | | |
| N6 | F6 | 25 | 26,28 | 27 | | | | |
| N7 | F | 31 | 32,34 | 33 | | | | |
| N8 | F8 | 35 | 36,38 | 37 | | | | |
| N9 | F9 | 19 | 20,40 | 39 | | | | |
| N10 | F1 Output pulse amplitude, low level | 2 | 29,3 | 4 | 0.0 | 0.4 | | |
| N11 | F2 | 12 | 9,7 | 8 | | | | |
| N12 | F3 | 18 | 17,5 | 6 | | | | |
| N13 | F4 | 16 | 15,13 | 14 | | | | |
| N14 | F5 | 21 | 22,24 | 23 | | | | |
| N15 | F6 | 25 | 26,28 | 27 | | | | |
| N16 | F7 | 31 | 32,34 | 33 | | | | |
| N17 | F8 | 35 | 36,38 | 37 | | | | |
| N18 | F9 | 19 | 20,40 | 39 | | | | |





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TABLE XIII. Transition time ($t_{THL}$ and $t_{TLH}$) test.

| Test condition | Parameter | Connect | | | Initial limits 25°±5°C ambient | | Units | Test figure |
|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | Min | Max | | |
| P1 | F1 Transition time ($t_{THL}$) | 2 | 3,29 | 4 | 6 | 24 | ns | 16,17 |
| P2 | F2 | 12 | 9,7 | 8 | | | | |
| P3 | F3 | 18 | 17,5 | 6 | | | | |
| P4 | F4 | 16 | 15,13 | 14 | | | | |
| P5 | F5 | 21 | 22,24 | 23 | | | | |
| P6 | F6 | 25 | 26,28 | 27 | | | | |
| P7 | F7 | 31 | 32,34 | 33 | | | | |
| P8 | F8 | 35 | 36,38 | 37 | | | | |
| P9 | F9 | 19 | 20,40 | 39 | | | | |
| P10 | F1 Transition time ($t_{TLH}$) | 2 | 3,29 | 4 | 60 | 160 | | |
| P11 | F2 | 12 | 9,7 | 8 | | | | |
| P12 | F3 | 18 | 17,5 | 6 | | | | |
| P13 | F4 | 16 | 15,13 | 14 | | | | |
| P14 | F5 | 21 | 22,24 | 23 | | | | |
| P15 | F6 | 25 | 26,28 | 27 | | | | |
| P16 | F7 | 31 | 32,34 | 33 | | | | |
| P17 | F8 | 35 | 36,38 | 37 | | | | |
| P18 | F9 | 19 | 20,40 | 39 | | | | |



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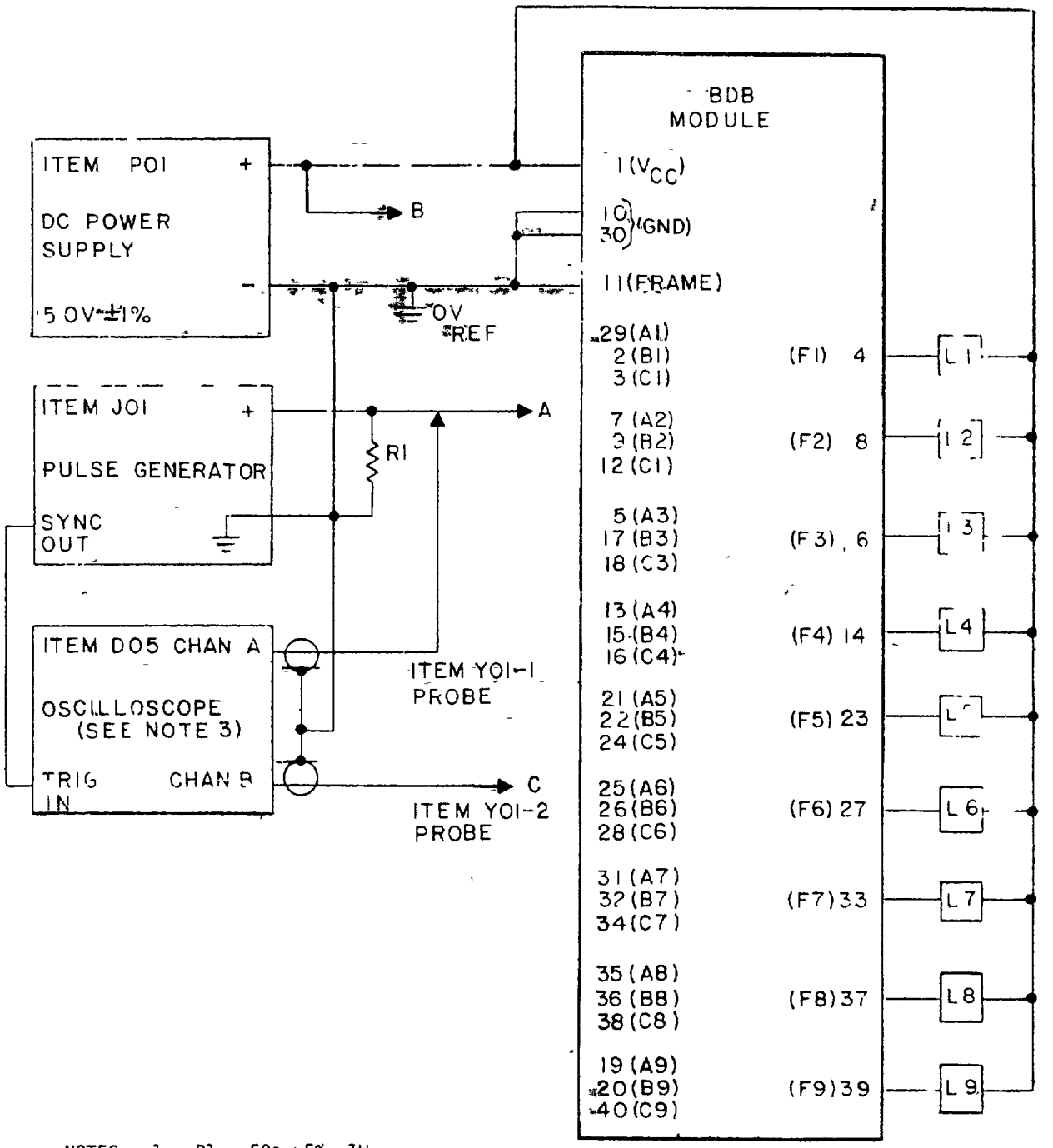
TABLE XIV. Propagation delay time ($t_{PLH}$ and $t_{PHL}$) test.

| Test condition | Parameter | Connect | | | Initial limits 25°±5°C ambient | | Units | Test figure |
|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | Min | Max | | |
| Q1 | F1 Propagation delay time ($t_{PLH}$) | 2 | 3,29 | 4 | 20 | 75 | ns | 16,17 |
| Q2 | F2 | 12 | 9,7 | 8 | | | | |
| Q3 | F3 | 18 | 17,5 | 6 | | | | |
| Q4 | F4 | 16 | 15,13 | 14 | | | | |
| Q5 | F5 | 21 | 22,24 | 23 | | | | |
| Q6 | F6 | 25 | 26,28 | 27 | | | | |
| Q7 | F7 | 31 | 32,34 | 33 | | | | |
| Q8 | F8 | 35 | 36,38 | 37 | | | | |
| Q9 | F9 | 19 | 20,40 | 39 | | | | |
| Q10 | F1 Propagation delay time ($t_{PHL}$) | 2 | 3,29 | 4 | 6 | 40 | | |
| Q11 | F2 | 12 | 9,7 | 8 | | | | |
| Q12 | F3 | 18 | 17,5 | 6 | | | | |
| Q13 | F4 | 16 | 15,13 | 14 | | | | |
| Q14 | F5 | 21 | 22,24 | 23 | | | | |
| Q15 | F6 | 25 | 26,28 | 27 | | | | |
| Q16 | F7 | 31 | 32,34 | 33 | | | | |
| Q17 | F8 | 35 | 36,38 | 37 | | | | |
| Q18 | F9 | 19 | 20,40 | 39 | | | | |





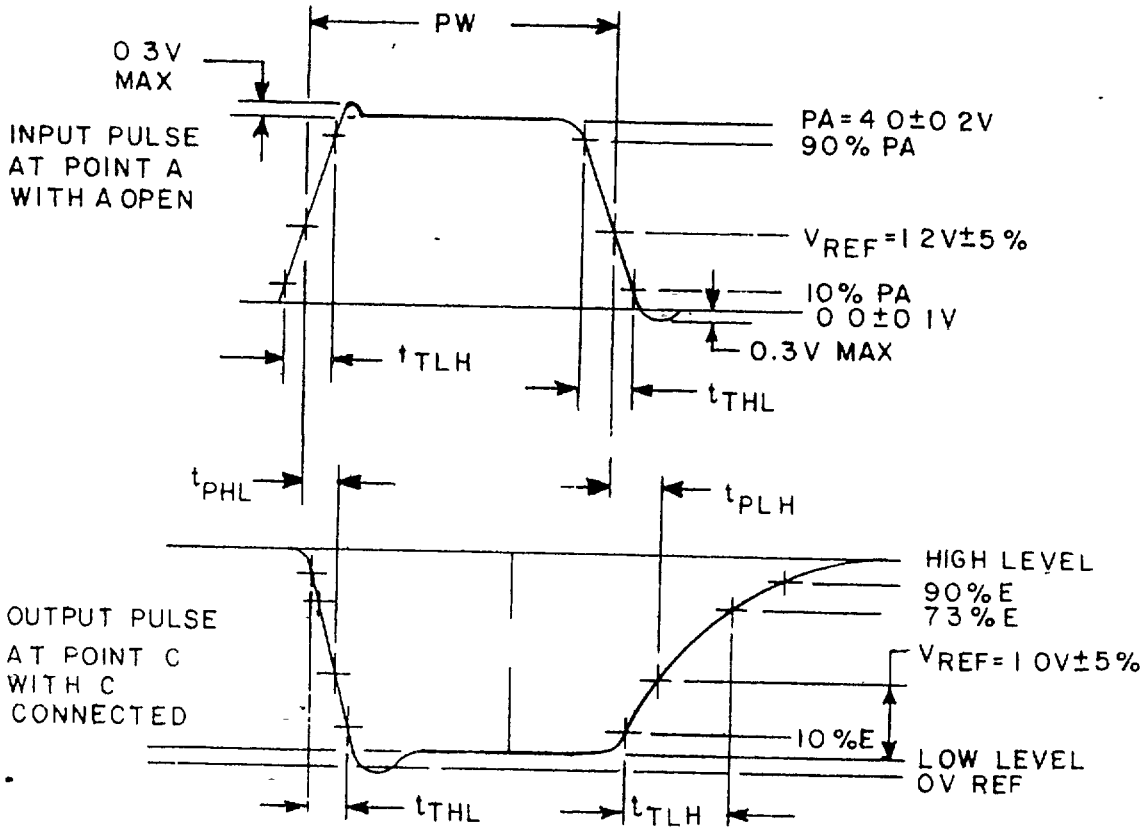
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NOTES:
1. R1 = 50Ω ±5%, 1W.
2. L1-L9 = Load circuit number 1 as shown on figure T5.
3. For test conditions N1-N14 use item D03 with item Y04 probe. The voltage measurements must have a 5 mV per millimeter resolution.

FIGURE 16. Output pulse amplitude (high-level and low-level), transition time ($t_{THL}$ and $t_{TLH}$), and propagation delay time ($t_{PLH}$ and $t_{PHL}$) test circuit for tables XII, XIII and XIV.

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NOTE: Input pulse parameters:
PW = 1 µs ±10%.
PRR = 100 kHz ±10%.
$t_{TLH} = t_{THL} = 20 \pm 2$ ns.

FIGURE 17. Output pulse amplitude (high-level and low-level), transition time ($t_{THL}$ and $t_{TLH}$), and propagation delay time ($t_{PLH}$ and $t_{PHL}$) test circuit waveforms for tables XII, XIII and XIV.

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TABLE XV. Output crosstalk (high-level and low-level) test.

| Test condition | Parameter | | Connect | | | | Initial limits 25°±5°C ambient (probe Y01-2) | | Units | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | J to points | K to point | M to point | N to pin | Min | Max | | |
| R1 | F1 | Output crosstalk, high level | B,C,D,E, F,G,H,I | A | Open | 4 | 0.0 | 0.8 | V | 18,19 |
| R2 | F2 | | A,C,D,E, F,G,H,I | B | | 8 | | | | |
| R3 | F3 | | A,B,D,E, F,G,H,I | C | | 6 | | | | |
| R4 | F4 | | A,B,C,E, F,G,H,I | D | | 14 | | | | |
| R5 | F5 | | A,B,C,D, F,G,H,I | E | | 23 | | | | |
| R6 | F6 | | A,B,C,D, E,G,H,I | F | | 27 | | | | |
| R7 | F7 | | A,B,C,D, E,F,H,I | G | | 33 | | | | |
| R8 | F8 | | A,B,C,D, E,F,G,I | H | | 37 | | | | |
| R9 | F9 | | A,B,C,D, E,F,G,H | I | | 39 | | | | |
| R10 | F1 | Output crosstalk, low level | B,C,D,E, F,G,H,I | Open | A | 4 | | | | |



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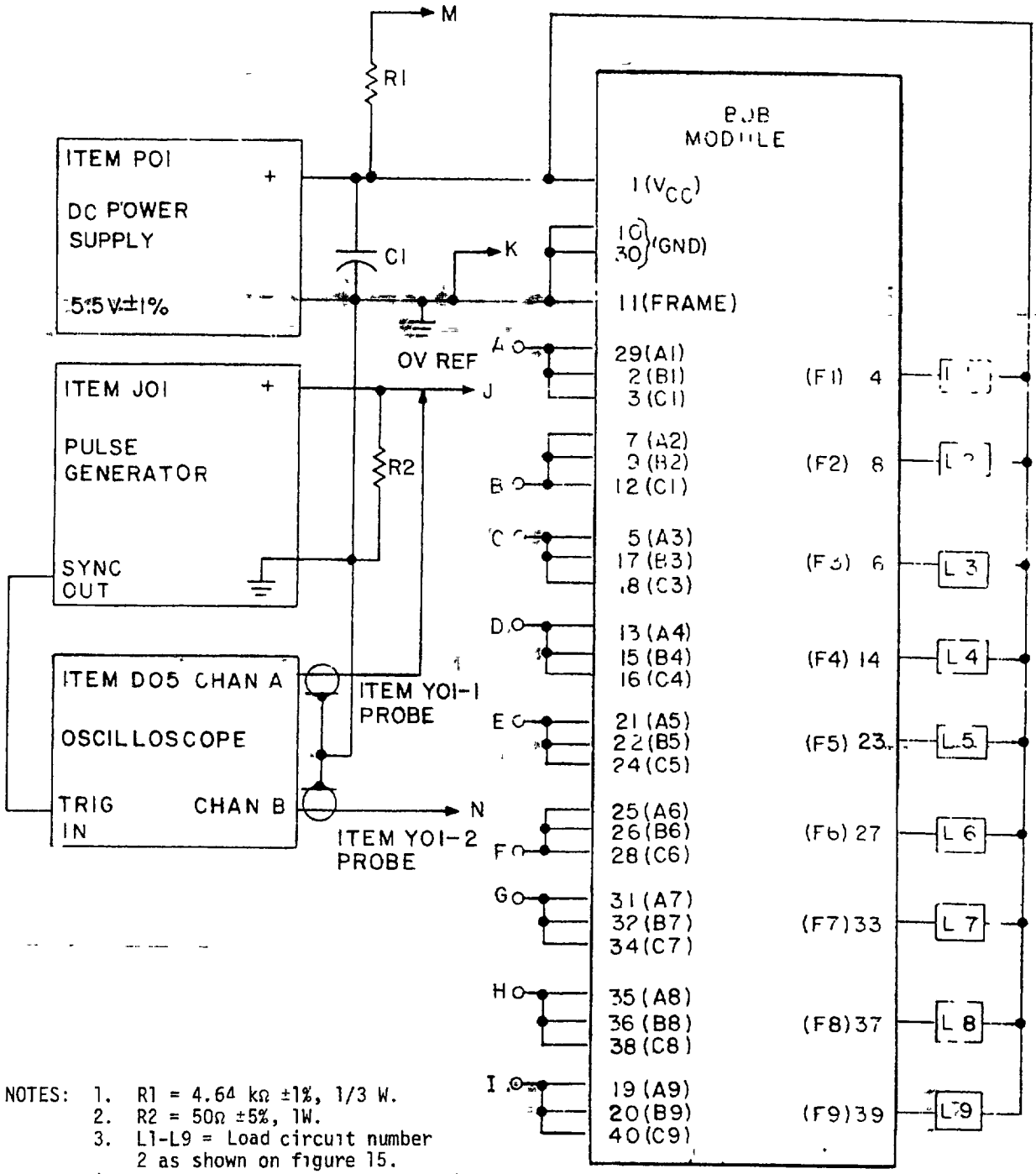
TABLE XV. Output crosstalk (high-level and low-level) test - Continued

| Test condition | Parameter | | Connect | | | | Initial limits 25°±5°C ambient (probe Y01-2) | | Units | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | J to points | K to point | M to point | N to pin | Min | Max | | |
| R11 | F2 | Output crosstalk, low level | A,C,D,E, F,G,H,I | Open | B | 8 | 0.0 | 0.8 | V | 18,19 |
| R12 | F3 | | A,B,D,E, F,G,H,I | | C | 6 | | | | |
| R13 | F4 | | A,B,C,E, F,G,H,I | | D | 14 | | | | |
| R14 | F5 | | A,B,C,D, F,G,H,I | | E | 23 | | | | |
| R15 | F6 | | A,B,C,D, E,G,H,I | | F | 27 | | | | |
| R16 | F7 | | A,B,C,D, E,F,H,I | | G | 33 | | | | |
| R17 | F8 | | A,B,C,D, E,F,G,I | | H | 37 | | | | |
| R18 | F9 | | A,B,C,D, E,F,G,H | | I | 39 | | | | |





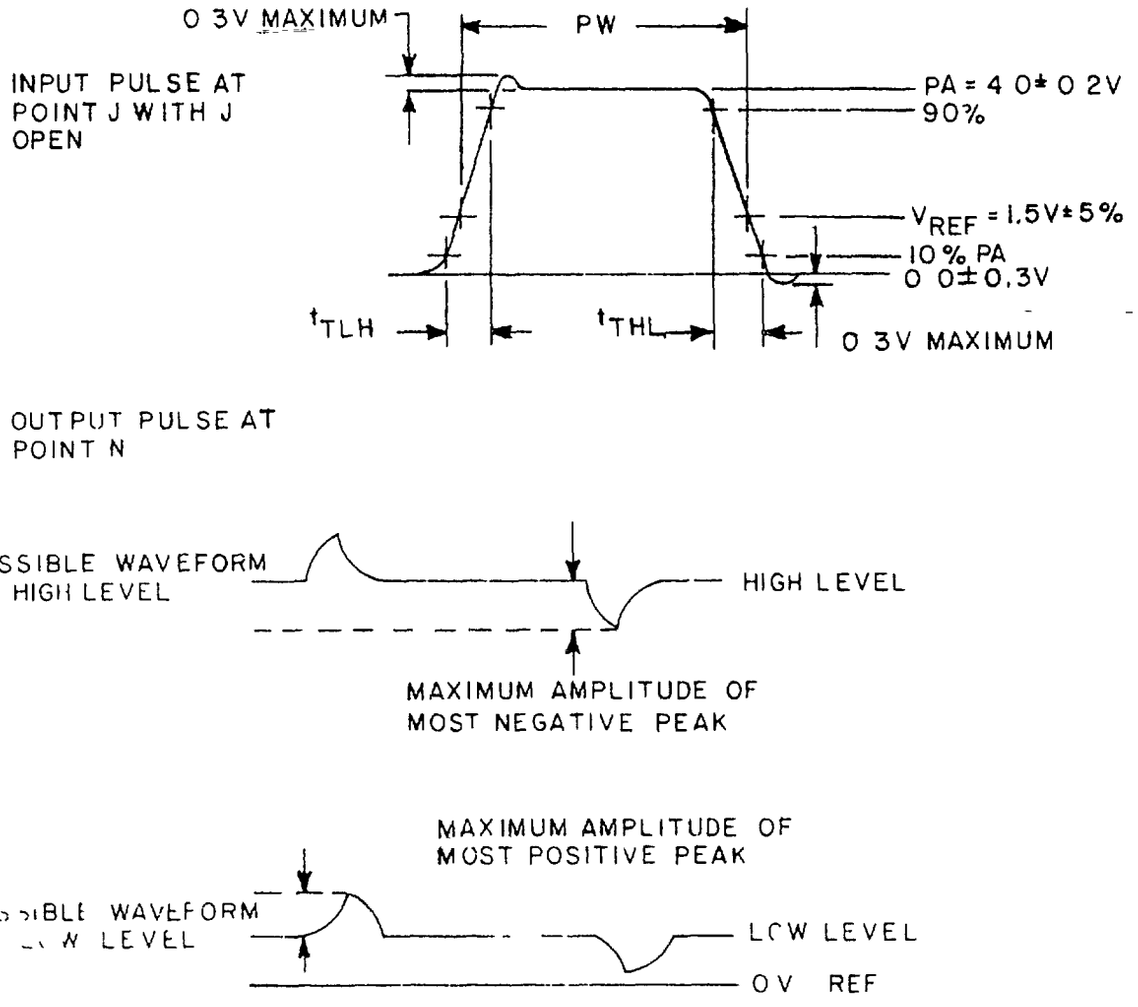
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NOTES:
1. R1 = 4.64 kΩ ±1%, 1/3 W.
2. R2 = 50Ω ±5%, 1W.
3. L1-L9 = Load circuit number 2 as shown on figure 15.
4. C1 = Represents power supply bypass capacitance added to the test fixture to limit the transient response of the power supply and the power supply transmission paths to module current demands. Capacitance may be added as required.

FIGURE 18. Output crosstalk (high-level and low-level) test circuit for table XV.

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NOTE: Input pulse parameters
PW = 0.9 ±0.1 µs.
$t_{TLH} = t_{THL}$ = 20 ±2 ns.
PRR = 100 kHz ±10%.

FIGURE 19. Output crosstalk (high-level and low-level) test circuit waveforms for table XV.



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